

Theorema Aureum 143 – v2.3

P5_genuine = 1000000001119 [13 digits, Sym=1] LOCKED

Source: docs/COLOPHON.md (verbatim). Generated 2026-06-02.

Colophon

****Title:** Towers YM v2.3 – H4 Boundary Protocol**
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****Version:** 2.3 FINAL**

Build protocol

****Formal + Empirical****
****Axioms:** 0**
****Sorry:** 0**
****Mathlib:** OFF (mathlib-free, Lean core only)**
****Provenance:** ON**

Modules

On-disk filenames use underscores; the published module names are listed first.

1. ****H4Core**** (`Towers/YM/H4Core.lean` + leaf `Towers/YM/H4\_Strata\_Ztau.lean`) – Task #324 – manifest SHA *record* `48536d9d...ba997` (prose milestone record hash, NOT a file hash). Real file SHA-256 `aa8c1180...a6ce`, recorded in `H4Core.data.json`.
2. ****H4Boundary**** (`Towers/YM/H4\_Boundary.lean`) – Task #325 – `C13 := 13`. ****Certified 13-character P5 table**** (digit counts tool-certified via `{#literal}`, never eye-counted):

Name	Value	Digits	Sym	Status
`P5_genuine`	`1000000001119`	13	1	**CERTIFIED · LOCKED**
`P5_proposed_14`	`10000000001119`	14	–	REJECTED
`P5_proposed_15`	`100000000001119`	15	–	REJECTED

`P5\_genuine = 1000000001119` is the ONLY live P5 in v2.3 – the real 13-digit boundary prime (`digit\_len = 13`, `Sym = 1`). The two over-zero variants each carry one extra zero, so their `digit\_len P5 = C13` (= 13) claim is ****FALSE****; they are kept solely as labeled ****REJECTED**** counter-records, not used as P5 anywhere. `H4\_Boundary.data.json` records the 14-digit `10000000001119` as the honest counter-record. No spike, transition, or `Sym`/critical-line meaning is attached to digit length (see `docs/BOUNDARY\_ARTIFACT.md`).

3. ****H4TimeBound**** (`Towers/YM/H4\_TimeBound.lean`) – Task #326 – `N40 := 40`; `TimeHorizon := 3^40` carries no causal/temporal meaning; `10^12 << 3^40` (by ~7 orders of magnitude).
4. ****H4Derivation**** (`Towers/YM/H4\_Derivation.lean`) – Task #327 – file SHA-256 `e21aelba...e2b2`. `C13\_Law\_Open`: ****CONJECTURE**** (named open `Prop`; no theorem discharges it).

Core results

****Primary law (CONJECTURE – empirical 6/6 only, NOT proven):****
`digit\_len p ≥ 13 → Sym p = 1`.

****Demoted:**** `N40 := 40`. `TimeHorizon := 3^40` is not causal – a plain integer the witnesses are compared against.

****Computed (real $W(H_4)$ action, not hardcoded):**** `Sym(19) = 2`, `Sym(191) = 2`,
`Sym(1000000001119) = 1`. (`Sym(19) = 2`, NOT 20.)

Gates

Gematria: OFF | Abjad: OFF | Religious: OFF | AI Attribution: OFF

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